# An Exact Inclusion–Exclusion Formula for the Prime-Counting Sequence

A note on OEIS A000720

**Primary author (AI): ChatGPT (OpenAI)**

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**Abstract**

We derive an exact finite formula for the prime-counting function $\pi(n)$ using sieving by primes up to $\sqrt{n}$. A corresponding recurrence counts integers avoiding the first $a$ primes. The argument is elementary, gives exact values rather than an asymptotic approximation, and is a classical form of Legendre's prime-counting method.

## 1 Prime counting from a finite sieve

OEIS A000720 records $\pi(n) = \#\{p \le n : p \text{ prime}\}$ for positive integers $n$. Write $p_1 = 2, p_2 = 3, \dots$ for the increasing primes and define

$$\Phi(n, a) := \#\{1 \le m \le n : p_j \nmid m \text{ for all } 1 \le j \le a\},$$

where $\Phi(n, 0) = n$ for $n \ge 0$. Let $P_a := \prod_{j=1}^{a} p_j$, with $P_0 = 1$.

**Theorem 1** (Exact finite inclusion–exclusion)**.** *For each integer $n \ge 1$, let $a := \pi(\lfloor \sqrt{n} \rfloor)$. Then*

$$\boxed{\pi(n) = a - 1 + \sum_{d \mid P_a} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor .} \tag{1}$$

*Equivalently,*

$$\pi(n) = a - 1 + \sum_{S \subseteq \{1,\dots,a\}} (-1)^{|S|} \left\lfloor \frac{n}{\prod_{j \in S} p_j} \right\rfloor . \tag{2}$$

*The formulas include $n = 1$ and the empty product convention.*

*Proof.* By inclusion–exclusion, the number of integers $m \in [1, n]$ divisible by none of $p_1, \dots, p_a$ is precisely

$$\Phi(n, a) = \sum_{d \mid P_a} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor ,$$

because every divisor of $P_a$ is squarefree. Any composite $m \le n$ has a prime divisor not exceeding $\sqrt{m} \le \sqrt{n}$, and is therefore removed by this sieve. Conversely, 1 and every prime $p > \sqrt{n}$ survive. Thus the survivors are exactly 1 and the $\pi(n) - a$ primes exceeding $\sqrt{n}$. Hence $\Phi(n, a) = 1 + \pi(n) - a$, proving both expressions. □

## 2 A recursive version and checks

**Proposition 1** (Sieving recurrence)**.** *For integers $n \geq 0$ and $a \geq 1$,*

$$\Phi(n, a) = \Phi(n, a-1) - \Phi\left(\left\lfloor \frac{n}{p_a} \right\rfloor, a-1\right), \quad \Phi(n, 0) = n. \tag{3}$$

*Proof.* Start with the integers not divisible by any of $p_1, \ldots, p_{a-1}$. Those newly removed by the factor $p_a$ are numbers of the form $p_a t$, where $1 \leq t \leq \lfloor n/p_a \rfloor$ and none of the earlier primes divides $t$. Their count is the subtracted term. This is valid also when $n < p_a$, in which case that term is zero. □

For $n = 30$, the sieve primes are $2, 3, 5$, so $a = 3$ and

$$\Phi(30, 3) = 30 - (15 + 10 + 6) + (5 + 3 + 2) - 1 = 8, \quad \pi(30) = 3 - 1 + 8 = 10.$$

For $n = 100$, $a = 4$ and $\Phi(100, 4) = 22$, whence $\pi(100) = 25$. Exact integer computation verified (1) for every $1 \leq n < 150$.

## 3 Scope and relation to open questions

Formula (1) is an exact combinatorial identity. It does not give the prime number theorem, new bounds for primes, or conclusions about the Riemann hypothesis. In particular, the identity must not be misrepresented as a new solution to a hard prime-distribution problem. It is a standard sieve observation, here written as a self-contained note for A000720.

## References


[1] OEIS Foundation, *A000720: $\pi(n)$, the number of primes $\leq n$*, https://oeis.org/A000720 (accessed November 13, 2026).

[2] R. Crandall and C. Pomerance, *Prime Numbers: A Computational Perspective*, Springer, 2001 (classical sieving and counting techniques).

[3] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976 (prime counting and elementary number-theoretic methods).